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Naylor, Robert Luke

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Chernobyl, Dark Waters and the contingency of environmental disaster and scientific knowledge

ROBERT LUKE NAYLOR
Centre for the History of Science, Technology and Medicine, University of Manchester; Department of Science Education, University of Copenhagen, Denmark

Abstract

This article contributes towards scholarship on teaching environmental history by analysing a series (*Chernobyl*, 2019) and film (*Dark Waters*, 2019). It argues that the immediacy of such media offer an ideal entry point into environmental history and history of science for students first studying those subjects. As well, it provides a model for how to use media in environment history.

Keywords: environmental history, film and environmental history, teaching

There is a nascent but growing literature on teaching environmental history. Emily Wakild and Michelle Berry provide a framework for constructing environmental history courses, advocating a teaching style that focuses on skill acquisition and embraces the rise of online communications through a 'cyborg pedagogy' of class polls, Twitter hashtags and collaborative editing.[1] Drawing on almost two decades of experience, Andrea Gaynor argues for the importance of environmental history teaching in the current era despite the subject's lack of recognition, and advocates for a more diverse and decolonised curriculum.[2] In a broad international survey of teaching in the environmental humanities, a large collaboration led by Emily O'Gorman argues that a core set of teaching approaches in the environmental humanities would be desirable, with the caveat that teaching in this area should remain 'an open, experimental, emergent space of possibilities'. Proposals for

1 Emily Wakild and Michelle K. Berry, *A Primer for Teaching Environmental History: Ten Design Principles* (Durham, NC: Duke University Press Books, 2018).

2 Andrea Gaynor, 'Environmental History: Teaching in a Time of Crisis', in *Teaching History for the Contemporary World: Tensions, Challenges and Classroom Experiences in Higher Education*, ed. Adele Nye and Jennifer Clark (Singapore: Springer, 2021), 171–81, doi.org/10.1007/978-981-16-0247-4_12.

features of this common approach include a rejection of the concept of a monolithic 'humanity', a rejection of an 'environment' separate from society, and a critical and situated approach to knowledge formation.[3]

Thus far, there is less focus in the literature on media formats of sources used in teaching undergraduate environmental history and their respective strengths and weaknesses. In this short article, I take inspiration from my experience teaching in late 2020 on the course 'Climate Change & Society' at the University of Manchester and argue that film and television dramatisations of environmental disasters can play an important role in helping students break down the barriers between environmental disaster, science and society. The course was run through both Manchester's Centre for the History of Science, Technology and Medicine and University College for Interdisciplinary Learning, which allowed students from almost any discipline to enrol. In practice, there was about an even split between humanities and science students, mostly in their third year of study, and just over 100 students in all. One course leader (Vladimir Janković) and three teaching assistants (including myself) ran the course, with all four members of the teaching staff splitting the marking of coursework and the running of weekly seminar discussion groups. Due to the pandemic, the course took place almost completely online. Although the course handout included a long list of films about climate change, the required readings were all in written form.[4]

Helping teach an interdisciplinary undergraduate course on climate change and society presents unique problems, depending on the student in question. On the one hand, based on my own experience, humanities students often lack confidence when using science papers as primary sources—a problem that the course leader tried to remedy by giving students a crash course in atmospheric science in an early lecture. On the other, science students often struggle to see, let alone form an opinion on, how scientific knowledge and environmental disaster are contingent upon the societies from which they arise. For many, scientific knowledge forms an 'oasis' outside the murk of economics and politics, a view that is often encouraged

3 Emily O'Gorman et al., 'Teaching the Environmental Humanities: International Perspectives and Practices', *Environmental Humanities* 11, no. 2 (November 2019): 427–60, doi.org/10.1215/22011919-7754545.

4 Examples include: Matthias Heymann, 'The Evolution of Climate Ideas and Knowledge', *WIREs Climate Change* 1, no. 4 (2010): 581–97, doi.org/10.1002/wcc.61; Martin Mahony and Mike Hulme, 'Modelling and the Nation: Institutionalising Climate Prediction in the UK, 1988–92', *Minerva* 54, no. 4 (December 2016): 445–70, doi.org/10.1007/s11024-016-9302-0; Daniel Sarewitz, 'Does Climate Change Knowledge Really Matter?', *WIREs Climate Change* 2, no. 4 (2011): 475–81, doi.org/10.1002/wcc.126; Tim Hayward, 'Climate Change and Ethics', *Nature Climate Change* 2, no. 12 (December 2012): 843–48, doi.org/10.1038/nclimate1615; Vladimir Janković and David M. Schultz, 'Atmosfear: Communicating the Effects of Climate Change on Extreme Weather', *Weather, Climate, and Society* 9, no. 1 (January 2017): 27–37, doi.org/10.1175/WCAS-D-16-0030.1; Erik Swyngedouw, 'Apocalypse Forever?', *Theory, Culture & Society* 27, no. 2–3 (March 2010): 213–32, doi.org/10.1177/0263276409358728; Harriet Bulkeley and Peter Newell, 'Governing Climate Change: A Brief History', in *Governing Climate Change*, 2nd ed. (Abingdon: Routledge, 2015); Ted Nordhaus, 'The Empty Radicalism of the Climate Apocalypse', *Issues in Science and Technology* 35, no. 4 (2019): 69–78.

by undergraduate science curriculums and popular media.[5] Equally, many students see environmental disasters as aberrations in political and economic systems, rather than a product of them. Overcoming these beliefs is usually a fundamental first step towards developing a critical perspective on the science/society interaction, a perspective that is sorely needed in these times of pandemic, climate change and renewed geopolitical conflict.

No matter how compelling they may be, immediately throwing philosophy of science papers at a science undergraduate is rarely an effective way of changing hearts and minds. This is especially true at a time when students largely lead a cyborg existence, where students are as (if not more) likely to gain information from social media, video games or Netflix as traditional media.[6] In my view, media that have a more human touch, such as television and film, are more likely to cut through preconceptions. Although movie tropes such as the 'lone genius' and the 'evil genius' help reinforce the oasis view of scientific knowledge and the aberration view of environmental disaster respectively (*The Imitation Game* and the James Bond franchise immediately spring to mind), a few productions in recent years have challenged the stereotypes. Two in particular, both concerning notable environmental disasters, stand out as instructive, partly because they disagree with each other in so many ways—the perfect breeding ground for critical analysis.

The smash hit 2019 television series *Chernobyl*, made sadly relevant again by the war in Ukraine, follows the tragic events following the 1986 disaster, masterfully building up tension and not letting go of its audience until all five episodes are done. From the very start, the production makes its claim clear that the Soviet system was to blame for the catastrophe. Shots of Lenin statues, Soviet mosaics and party badges illustrate squabbling bureaucrats as they attempt to shift blame to one another rather than grappling with the real problems. As the series progresses, a group of enterprising scientists wade against the system to uncover the 'truth': that the disaster was caused by a state-induced performative culture that covered up safety concerns. Although it is important to remember that serious nuclear accidents were and are not restricted to the Soviet style of government, *Chernobyl* is brilliant at making the general point that environmental disaster cannot be separated from political systems. 'We did everything right' says one of the dying control engineers. Indeed, Chernobyl was not caused by chance or the incorrect press of a button, but by a multitude of systemic failures. I wish this message came across from cinema more. There was no evil genius who blew up Chernobyl as would be the case in a Bond film, rather a number of flawed individuals within a flawed system.[7]

5 Robert Luke Naylor, 'How (Not) to Build an Expert: Personal Reflections on a Terminated Physics Career', *Spontaneous Generations: A Journal for the History and Philosophy of Science* 10, no. 1 (2022): 98–106.

6 Wakild and Berry, *A Primer for Teaching Environmental History*, Chapter 9.

7 An HBO series written by Craig Mazin and directed by Johan Renck; total running time 5 hours 3 minutes.

However, although *Chernobyl* is relentless in placing the blame of environmental disaster on the societal system from which it emerged, it falls for standard tropes in its portrayal of science. Physical scientists are depicted as members of a secret society somehow immune to the influence of the party, serving as a proxy for the Western audience peering into the shortcomings of the system. 'Forgive me—maybe I've just spent too much time in my lab, or maybe I'm just stupid. Is this really the way it all works? An uninformed, arbitrary decision that will cost who knows how many lives that is made by some apparatchik, some career party man?,' asks the main character Valery Legasov, a career party man whose real-world status in the Soviet Union most certainly required some good apparatchik-ing. His co-conspirator, a fictional scientist by the name of Ulana Khomyuk, is a Western-relatable good capitalist citizen who comes into work on weekends while everyone else is having fun. Again, she seems blissfully unaware of the system she grew up in, spending most of her time taking on the trope of a Western spy. Perhaps the most unbelievable character is Boris Shcherbina, a seasoned bureaucrat who slowly gets converted to the scientist's pursuit of truth. 'It's beautiful,' the gruff man says at the sight of a caterpillar, as if seeing it for the first time, as the shackles of the system fall away and he embraces the purity of science. Although *Chernobyl* does a good job of highlighting the socially embedded nature of the disaster it portrays, it does a terrible job at highlighting that scientists are human beings conscious of how to interact with their societal system. As any science administrator knows (but might not say), science is as integrated into sociopolitical systems as any other human activity.

Inspired by a 2016 *New York Times* article,[8] the 2019 film *Dark Waters* goes some way to tackling *Chernobyl*'s artificial divide between science and society, thoroughly polluting the scientific oasis. *Dark Waters* follows the story of corporate defence lawyer Robert Bilott, who attempts to hold his former masters to account for creeping environmental disasters they knowingly caused in the name of profit. Some reviewers have denoted *Dark Waters* as America's *Chernobyl*,[9] but I think this does *Dark Waters* a disservice. It presents a much more nuanced and evocative portrayal of how a societal system not only causes environmental disaster but also bends scientific knowledge to its will. From the start, we see how scientific knowledge, institutions and practice are used by the DuPont corporation to avoid responsibility. DuPont-influenced Environmental Protection Agency reports are used to deflect blame from the company onto its victims, unexplained acronyms are used to hide the identity of dangerous substances from the scientifically uninitiated, and legitimate

8 Nathaniel Rich, 'The Lawyer Who Became DuPont's Worst Nightmare', *New York Times*, 6 January 2016, www.nytimes.com/2016/01/10/magazine/the-lawyer-who-became-duponts-worst-nightmare.html, accessed 29 June 2022.

9 'Review: Ominous "Dark Waters" is like "Chernobyl" for America', News, TribLIVE.com, 27 November 2019, triblive.com/aande/movies-tv/review-ominous-dark-waters-is-like-chernobyl-for-america, accessed 29 June 2022; Tom Leonard, 'Chernobyl in the USA: Thrilling new movie reveals how a heroic lawyer found the US giant behind Teflon was pumping out a deadly molecule that causes cancer and deformities for YEARS—and it's in 90 per cent of us', *Mail Online*, 20 February 2020, www.dailymail.co.uk/news/article-8026587/New-movie-reveals-lawyer-giant-Teflon-pumping-deadly-molecule.html, accessed 29 June 2022.

members of the scientific community work with DuPont to change the rules in order to sidestep a lawsuit. Meanwhile, decades-old DuPont-sponsored studies into the dangers of the chemicals it produces are buried. Control of scientific knowledge is used as a shield by those in positions of power: '[E]very scientist who knows anything about any of this already works for those chemical companies. That's no accident,' explains Bilott during a particularly tense exchange.

The closing credits of *Chernobyl* pontificate that although the individuals responsible for the disaster got off lightly, the literal and metaphorical fallout contributed to the fall of the Soviet system itself. In contrast, for any Western audience of *Dark Waters*, it is not some alien authoritarian system from history that caused the deaths and suffering of innocent people, but our own, very present, system of societal organisation and motivation. The individual perpetrators of the disasters portrayed in *Dark Waters* faced no consequences at all and DuPont remains a titan of American industry. *Dark Waters* oozes with injustice, highlighting the asymmetry in a legal system where restitution is usually monetary, as well as the bravery of those who decide to make a stand: 'Don't want no money. The whole damn world need to see what they done!' exclaims Wilbur Tennant, a farmer made destitute by the actions of DuPont, angry that no one would face imprisonment for the crimes committed against him. *Dark Waters* is a film about oppression and, in this case, science is a tool of the oppressor.

I believe that it is important to engage students with such stark critiques. In my view, productions such as *Dark Waters* and *Chernobyl* should form part of the curriculum for undergraduate courses in environmental history, especially when teaching students from across a range of disciplines. Analysing film and television allows science students to begin to critically analyse without needing to immediately tackle intimidating humanities papers—it forms an easy entry point to thinking critically about the science/society interaction in seminars and lectures. Productions such as these also induce a visceral, emotional reaction in viewers, perhaps helping students of science overcome the institutionally imposed mental hurdles that often stop them from stating their own opinions and disagreeing with authority in academia.[10] Both productions have been criticised for inaccuracies (*Dark Waters* by DuPont itself no less),[11] but for teaching purposes this does not detract from their value, only adding more avenues of analysis and discussion. They present compelling stories that help

10 Naylor, 'How (Not) to Build an Expert'.

11 Masha Gessen, 'What HBO's "Chernobyl" Got Right, and What It Got Terribly Wrong', *New Yorker*, 4 June 2019, www.newyorker.com/news/our-columnists/what-hbos-chernobyl-got-right-and-what-it-got-terribly-wrong, accessed 29 June 2022; Jim Smith, 'Ten times the Chernobyl television series lets artistic licence get in the way of facts', *The Conversation*, 22 June 2019, theconversation.com/ten-times-the-chernobyl-television-series-lets-artistic-licence-get-in-the-way-of-facts-119110, accessed 29 June 2022; Mark Eichmann, 'DuPont execs react to villain role in "Dark Waters" film', *WHYY* (blog), 1 November 2019, whyy.org/articles/dupont-execs-react-to-villain-role-in-dark-waters-film, accessed 29 June 2022.

integrate environmental disaster and scientific practice into our societal systems, an integration that is essential if we are to see any meaningful societal changes to meet the challenges of today.

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